

A - Structured Programming

Task 1 : Write a simple C++ program that declares and initializes an array of values. The program will then use the for loop to display the output. Use the for-each loop to prompt user for values and store the values back into the array.

Sequence :

- 1 – Declare an array with values
- 2 – Display values in the array
- 3 – Use enhanced for loop to read and store values back into the array
- 4 – Display the new values in the array

Task 2 : Write a simple C++ program that declares and initializes an array of integer values. The program will then use the for loop to display the values in the array. Use the for-each loop to change the even numbers from the list to 999. Display all the values in the array including the new values.

B – Procedural Programming (Procedure, Subroutines, Methods, Functions)

Task 3: Based on the sample codes given, modify the for loop within the functions – display() and readInput() using the enhanced for loop. What is your output / conclusion?
If there is an error, how do you rectify if you need to still use functions + for each loop?

C – Getting organised I – Structure within Structure, Functions

Task 4 : Write a simple C++ program that has the structure definition for **Address – string hseNo, block, condoName** and **Person – int pld, pAge, string pName, Address add**. Within the main function,

1. Declare a single variable, *me*, of **Person** and initialize your own data for **Person**, which includes the address
2. Declare a single variable, *friendAP*, of **Person**. Prompt user for the data for and store the data in the *friendAP*.
3. Declare an array variable of **Person** of SIZE 3. Prompt user for the data for and store the data in the array variable.
4. Write the C++ statements to display all the values in the variables in 1, 2 and 3. Additional: use the enhanced for loop to display the values in the array variable in 3{h

Task 5 : Modify the structure definition in **Task 4** to include a function called **setDetails** that accepts values for **Person** and a function called **displayPerson** that displays all values. If possible do not repeat codes. Write the function prototypes within the structures. Write the function definition outside the structure.

Task 6 : Modify and separate the codes in **Task 5** into 3 files – driver program, header file and implementation file.

Getting organised II: File Handling

Task 4 : Create a text file with 3 person records. Load the data into the array of **Person**. Allow the user to edit the record and save into new text file.